# Scott Powers

saberpowers.com

## Academic Experience

**Rice University** 2023–
Assistant Professor, Sport Analytics, (by courtesy) Statistics 2023–
Director, Hutchinson Leadership Initiative in Sport Analytics 2026–

## Sports Experience

### Full-Time

**Houston Astros** 2022
Assistant General Manager

**Los Angeles Dodgers** 2017–2021
Director, Quantitative Analysis 2019–2021
Senior Analyst, Research & Development 2017–2018

### Part-Time/Consulting

**Teamworks** (formerly Zelus Analytics) 2020, 2023–2024
Staff Data Scientist 2023–2024
Senior Data Scientist 2020

**AZ Alkmaar** 2015–2020
Data Analyst

**Oakland Athletics** 2015–2016
Analytics Consultant

## Education

**Stanford University** 2017
Ph.D. Statistics
Advisor: Rob Tibshirani

**University of Chicago** 2013
M.S. Statistics

**University of North Carolina at Chapel Hill** 2011
B.S. Mathematical Decision Sciences and Mathematics
*Highest Honors from the Department and Highest Distinction from the University*

## Research

### Working Papers

**Powers**, Graves R, Vannucci M, Buffi JH, Barrueco D, D'Angelo J, Glazer AK
"A large-scale study of pitch-tracking risk factors for professional baseball pitcher injuries"

**Powers**, Ramani S, Hahn J, Schaefer AJ
"Lead distance under a pickoff limit in Major League Baseball: A sequential game model"
*arXiv:2601.15608*

Zhu R, Love D, **Powers**
"Ball path curvature and in-game free throw shooting proficiency in the National Basketball Association"
*arXiv:2507.01238*

**Powers**, Iglesias V
"Pitch trajectory density estimation for predicting future outcomes"

## Journal Articles

Millington E, **Powers** (2026)
"Models in the War Room: Translating random forest predictions into player comps"
to appear in *Journal of Sports Analytics*

**Powers**, Yurko R (2026)
"Swinging, Fast and Slow: Interpreting variation in baseball swing tracking metrics"
to appear in *The American Statistician*

**Powers**, Stancil L, Consiglio N (2025)
"Estimating individual contributions to team success in women's college volleyball"
*Journal of Quantitative Analysis in Sports* **21**(2) 117–135

Burton T, **Powers**, Burns C, Conway G, Leach F, Senecal K (2023)
"A data-driven greenhouse gas emission rate analysis for vehicle comparisons"
*SAE International Journal of Electrified Vehicles* **12**(1)

**Powers**, McGuire V, Bernstein L, Canchola AJ, Whittemore AS (2019) "Evaluating disease prediction models using a cohort whose covariate distribution differs from that of the target population"
*Statistical Methods in Medical Research* **28**(1) 309–320

**Powers**, Hastie T, Tibshirani R (2018)
"Nuclear penalized multinomial regression with an application to predicting at bat outcomes in baseball"
*Statistical Modelling* **18**(5–6) 388–410

**Powers**, Qian J, Jung K, Schuler A, Shah NH, Hastie T, Tibshirani R (2018)
"Some methods for heterogeneous treatment effect estimation in high dimensions"
*Statistics in Medicine* **37**(11) 1767–1787

McGinnis L, **Powers**, Bangs D, Cherry A, Tibshirani R, Natkunam Y (2016)
"Double-hit diffuse large B-cell lymphomas with MYC gene rearrangements more commonly involve BCL2 than BCL6 gene rearrangements as the second hit: A large scale single institution study"
*Laboratory Investigation* **96** 362A

Burton T, **Powers** (2015)
"A linear model for estimating optimal service fraction in volleyball"
*Journal of Quantitative Analysis in Sports* **11** 117–129

**Powers**, Hastie T, Tibshirani R (2015)
"Customized training with an application to mass spectrometric imaging of cancer tissue"
*The Annals of Applied Statistics* **9** 1709–1725

**Powers**, DeJongh M, Best AA, Tintle NL (2015)
"Cautions about the reliability of pairwise gene correlations based on expression data"
*Frontiers in Microbiology* **6** 650

**Powers**, Gopalakrishnan S, Tintle NL (2011)
"Assessing the impact of non-differential genotyping errors on rare variant tests of association"
*Human Heredity* **72** 153–160

Saavedra S, **Powers**, McCotter T, Porter MA, Mucha PJ (2010)
"Mutually-antagonistic interactions in baseball networks"
*Physica A: Statistical Mechanics and its Applications* **389** 1131–1141

## Invited Conference Talks

"Safer, Not Slower: Actionable insights from a large-scale survival analysis of baseball pitching mechanics"
Joint Statistical Meetings — Boston 2026

"Winning baseball games by solving statistics puzzles"
International Conference on Statistics and Data Science — Vancouver 2025

"Estimating individual contributions to team success in women's college volleyball"
MIT Sloan Sports Analytics Conference — Boston 2024
New England Symposium on Statistics in Sports — Boston 2023

## Invited Department Seminars

"Winning baseball games by solving statistics puzzles"
Northeastern University, Network Science Institute — Mar 2026
Texas A&M University, Institute of Data Science — Feb 2026
Southern Methodist University, Department of Statistics and Data Science — Sep 2025
Johns Hopkins University, Department of Electrical and Computer Engineering — Sep 2025

"Player evaluation based on game performance"
New York Knicks (National Basketball Association) — Nov 2024
TSG 1899 Hoffenheim (Bundesliga) — Apr 2024

"Nuclear penalized multinomial regression"
Rice University, Department of Statistics — Apr 2023

## Invited Conference Panels

"Building the Academic Playbook for Sports Analytics Education"
MIT Sloan Sports Analytics Conference — Boston 2026

"The Past, Present, and Future of Sports Analytics in Industry and the Academy"
Joint Statistical Meetings — Nashville 2025

"Data Science and Analytics in Sports Performance and Management"
Columbia Symposium on AI and Sports — New York 2024

## Contributed Conference Talks

"Safer, Not Slower: Actionable insights from a large-scale survival analysis of baseball pitching mechanics"
Saberseminar — Chicago 2026

"Swinging, Fast and Slow: Untangling intention and timing error from bat speed and swing length"
Cascadia Symposium on Statistics in Sports — Vancouver 2024
Saberseminar — Chicago 2024

"Pitch trajectory density estimation for predicting future outcomes"
Conference of Texas Statisticians — Houston 2024
Saberseminar — Chicago 2023

"Jointly predicting exit velocity and launch angle for batter-pitcher matchups"
Saberseminar — Boston 2016

"Nuclear penalized multinomial regression for predicting at bat outcomes in baseball"
Joint Statistical Meetings — Chicago 2016

"True wOBA: Estimation of true talent level for batters"
SABR Analytics Conference — Phoenix 2016

"Rewarding batters for baserunner advancement: A ridge-regressed Rasch model"
Saberseminar — Boston 2015

## Student Conference Talks

"Bayesian hierarchical change point model for pitcher biomechanics"
Rose Graves @ Joint Statistical Meetings — Boston 2026

"A comps-based approach for interpreting tree-based predictions with an application to the NFL draft"
Elisabeth Millington @ Carnegie Mellon Sports Analytics Conference — Pittsburgh 2025

"Ball path curvature and in-game free throw shooting proficiency in the National Basketball Association"
Ruoqian (Judy) Zhu @ New England Symposium on Statistics in Sports — Boston 2025

"The Two-Foot Rule: A game theoretic analysis of the pickoff limit in Major League Baseball"
Jacob Hahn @ Cascadia Symposium on Statistics in Sports — Vancouver 2024
Jacob Hahn @ Saberseminar — Chicago 2024

"Modifying $k$-means clustering to optimize positioning in volleyball"
Naomi Consiglio @ Women in Sports Data Symposium — Philadelphia 2024

"Do we learn more about AAA batters when they face better pitches?"
Jeff Brover @ Saberseminar — Chicago 2024

"Time warping for clustering pitcher deliveries"
Drew Haugen @ Saberseminar — Chicago 2024

## Student Conference Posters

"A comps-based approach for interpreting tree-based predictions with an application to the NFL draft"
Elisabeth Millington @ New England Symposium on Statistics in Sports — Boston 2025

"A statistical approach to sport climbing difficulty and progression"
Elizabeth Sepúvelda @ Cascadia Symposium on Statistics in Sports — Vancouver 2024

"Not All Features Are Created Equal: Player clustering and evaluation"

# Funding

## External

**Major League Baseball Research Gift**
"Identifying Biomechanical Risk Factors for Pitcher Injuries" PI: S Powers, $100,000 2024–2026

**Major League Baseball Research Gift**
"Assessing Associations between Longitudinal Movement Screens and Performance in Baseball Pitchers"
PI: S Powers, $15,000 2026

## Internal

**Houston Methodist-Rice University Center for Human Performance Seed Grant Program**
"Utilizing Ultrasound-Based Shear Wave Elastography to Assess UCL Changes in Collegiate Pitchers"
PIs: S Powers, R Jack, $46,483 2026

**Rice University Creative Ventures Conference and Workshop Development**
"Rice Soccer Analytics Conference" PI: S Powers, $10,000 2024–2025

# Teaching

* indicates a course I designed

**Rice University**
Instructor, *SMGT 430: Introduction to Sport Analytics Spring 2024, 2026; Fall 2024, 2025
Instructor, *SMGT 435: Baseball Analytics Spring 2024; Fall 2024, 2025
Instructor, STAT 699: Topics in Statistical Sciences Spring 2026; Fall 2024, 2025
Instructor, *SMGT 432: Soccer Analytics Fall 2023
Instructor, *SMGT 434: Football Analytics Fall 2025
Instructor, SMGT 373: Sport Analytics Internship Spring 2026

**Stanford University**
Instructor, *STATS 50: Mathematics of Sports Spring 2016
TA, STATS 305A: Applied Statistics I Fall 2016
TA, STATS 216: Introduction to Statistical Learning Winter 2014, Summer 2015, Winter 2017
TA, STATS 202: Data Mining and Analysis Fall 2013, Summer 2014
TA, STATS 50: Mathematics of Sports Fall 2014

**University of Chicago**
Instructor, STAT 23400: Statistical Models and Methods Spring 2013
TA, STAT 22000: Statistical Methods and Applications Fall 2011, Spring 2012, Fall 2012

# Service

**To the Profession**
Lead Organizer, American Soccer Insights Summit 2024–
Sports Column Co-Editor, *CHANCE* 2026–
Associate Editor, *Journal of Quantitative Analysis in Sports* 2026–
Associate Editor, *Journal of Statistics and Data Science in Sports* 2026–
Associate Editor, SCORE Network 2024–
Organizer, ASA Virtual Sports Analytics Conference 2026

**Rice University**

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|---|---|
| Faculty Associate, Wiess College | 2024– |
| Instructor, STaRT@Rice (School of Social Sciences) | 2024, 2025 |
| Instructor, Owl Days Classroom Sampler (Office of Admission) | 2024 |
| Speaker, Special Occasion Dinner (Friends of Fondren Library) | 2024 |
| Speaker, Brains in a Bar (Association of Rice Alumni) | 2024 |
| Member, University Committee for Faculty and Staff Benefits | 2024 |

**Peer Review**

*Journal of Quantitative Analysis in Sports* (8), *Journal of Sports Analytics* (2), *International Journal of Sports Science & Coaching* (2), *Journal of Sport Management*, *Orthopaedic Journal of Sports Medicine*, *Journal of Sports Sciences*, *Experimental Physiology*, *Computer Methods in Biomechanics and Biomedical Engineering*, *Proceedings of the National Academy of Sciences*

## Professional Development

**Rice Center for Teaching Excellence**

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| INSTILL Mini-Grant Program | 2023–2025 |
| Reading Group | 2023–2025 |
| Symposium on Teaching and Learning | 2024 |
| Workshop on Inclusive Teaching Strategies | 2023 |

**Conferences Attended**

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|---|---|
| Women in Sports Data Symposium | Philadelphia 2023, 2024 |
| Minority Trailblazers in Sports Conference | Houston 2023 |

## Awards

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| Outstanding Associate, Wiess College | 2026 |
| Graduate Research Fellowship, National Science Foundation | 2011 |
| McCormick Fellowship, University of Chicago | 2011 |
| Mathematical Decision Sciences Award, UNC Department of Statistics & Operations Research | 2011 |
| Student Speaker Award, Pi Mu Epsilon National Meetings | 2010 |
| National Merit Scholarship, National Merit Scholarship Corporation | 2007 |
| Jack Kavanagh Memorial Youth Baseball Research Award, SABR | 2005, 2006, 2007 |

## Activities

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|---|---|
| Sports Analytics Club, Stanford University | 2014–2017 |
| Co-President | 2015–2016 |
| Intercollegiate Club Baseball, Stanford University | 2013–2015 |
| Intercollegiate Club Volleyball, Stanford University | 2013–2016 |
| Intercollegiate Club Volleyball, University of Chicago | 2011–2013 |
| Intercollegiate Club Volleyball, University of North Carolina at Chapel Hill | 2007–2011 |
| President | 2008–2010 |